

FARIHA SHAH

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Summary

Recent MSDS graduate from Seattle University, with 4+ years of experience building scalable ETL pipelines and cloud-native data systems using Python, SQL, and AWS. Experienced with AI-assisted development tools (Claude API, GitHub Copilot)

Experience

Data Engineering Consultant

Jan 2026 – Jun 2026

Seattle University × Costco Wholesale

- Designed and deployed an end-to-end ELT pricing analytics pipeline on AWS (Lambda, S3, EventBridge, SageMaker) to ingest and analyze competitive pricing data across Costco, Walmart, and Kroger
- Built log-log OLS cross-price elasticity models across 3 product categories; identified paper towel elasticity of **5.52** and up to **31%** projected volume lift through constrained price optimization
- Standardized **197K+** multi-retailer SKUs with automated data quality checks, normalizing price per unit for accurate cross-retailer comparison at scale

Graduate Data Assistant

Oct 2025 – June 2026

Division of Mission Integration, Seattle University, Seattle, WA

- Automated **5+** event data workflows using Power Automate, reducing manual coordination time by **60%** and compressing stakeholder response cycles from daily to hourly.
- Built SharePoint event database tracking **10K+ rows**; integrated with **Tableau** to deliver leadership dashboards, improving data accuracy by **30%**.

Data Engineer II

Jul 2020 – Aug 2024

FlowHCM, Karachi, Pakistan

- Diagnosed critical HR pipeline failure across 6 clients; redesigned ingestion layer, restoring executive reporting within **24 hours** and eliminating **10+ hrs/week** of manual reconciliation.
- Architected HR KPI platform integrating **50+ sources** processing **30M+ rows/month** using Python, PostgreSQL, C#; reduced analyst effort by **50%**; adopted as org-wide reporting standard.
- Re-engineered **12 ETL pipelines** on SQL Server processing **50M+ HR records**; cut runtimes by **40%** and infrastructure costs by **25%** with zero data loss.

Data Analyst Intern

Jan 2020 – Jun 2020

Pakistan Petroleum Limited, Karachi, Pakistan

- Built 5 ETL pipelines on geological datasets achieving **98% data integrity**; applied time-series forecasting using Python and **AWS SageMaker**, cutting reporting time by **15 hrs/week**.

Projects

Automated Data Pipeline with AI-Driven Anomaly Explanation

- Designed an LLM-powered pipeline on AWS processing **100K+ records**; engineered structured prompt templates generating context-aware anomaly explanations via **Claude API**, reducing manual debugging by **60%**.

Unstructured Data Ingestion & Semantic Search Engine

- Architected an AWS ingestion pipeline (Lambda, S3, Bedrock Titan Embeddings) processing **500+** daily documents into **10K+** pgvector vectors on RDS; exposed a FastAPI semantic search endpoint, replacing manual document retrieval entirely.

Retail Price Forecasting Pipeline

- Scraped **3.8M+ daily grocery prices** across **57 WA locations**; engineered **14-day forecasts** using Facebook Prophet (**MAPE < 4%**); delivered results on a live Streamlit dashboard.

Sentiment Classifier (DistilBERT Fine-Tuned on Amazon Reviews)

- Fine-tuned DistilBERT on **6,000 Amazon reviews** achieving **90.2% accuracy** and **0.902 F1**; deployed real-time predictions via FastAPI REST endpoint with **98.3%** inference confidence.

Education

M.S. in Data Science

June 2026

Seattle University, Seattle, WA

B.E. in Petroleum & Natural Gas Engineering

2016 – 2019

Dawood University of Engineering & Technology, Pakistan

Skills

Languages: Python, SQL, R, C#

AI & ML: scikit-learn, TensorFlow, PyTorch, Time-Series Forecasting, A/B Testing, Predictive Modeling

Cloud & Data Infra: Lambda, S3, EMR, Redshift, Glue, Athena, DynamoDB, EventBridge, SageMaker, Bedrock, QuickSight, Docker, Snowflake

ETL & BI Tools: Data Modeling, Data Warehousing, REST APIs, DBT, Kafka, PySpark, Apache Spark, Hadoop, Tableau, Power BI, Power Automate, Excel

Dev Tools: Git, Bash, UNIX, GitHub Actions, GitHub Copilot, Claude API